

Supporting Appendix: Full Version

The Foreign Policy Impact of Trade-Based Status Gains: When China Overtakes the US as
Top Export Destination

Prepared for Chinese Political Science Review

This file contains the complete supporting appendix. It is the version to use when the journal does not impose an appendix-length constraint.

Included sections

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Brazil and China Ideal-Point Series

Figure 7 plots the separate UNGA ideal-point series for Brazil and China. This diagnostic clarifies the interpretation of the absolute-distance outcome used in the main SDiD design: China's average position changes little around 2009 despite sizable year-to-year volatility, while Brazil shows a sustained post-2009 movement toward China.

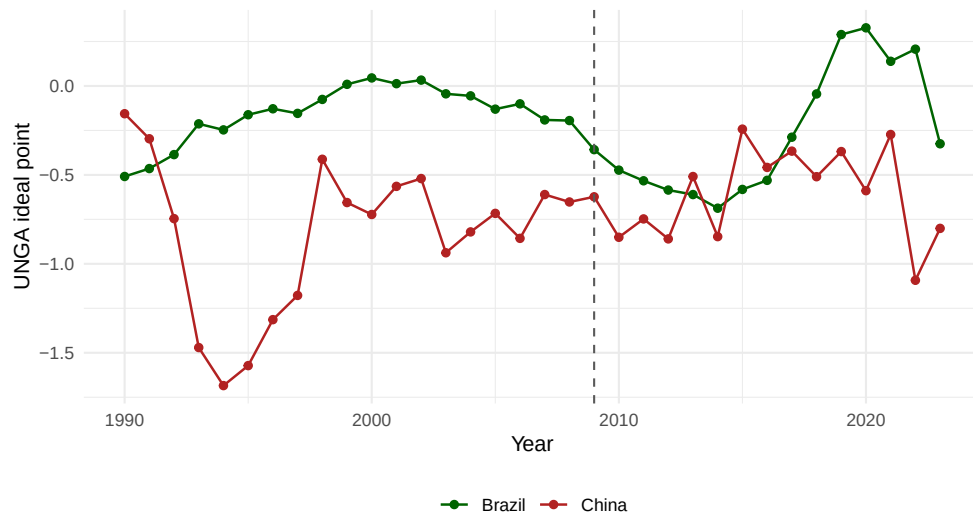


Fig. 7 Brazil and China UNGA ideal-point series. The dashed line marks 2009. The figure separates the two series behind the absolute-distance outcome used in the main SDiD analysis. Note: UNGA ideal-point score (Bailey et al. scale). Window: 1990-2022 annual series.

Triple-Difference Pre-Trends

Figure 8 reports the input to the identification assumption behind the triple difference. Each series is Brazil's average distance to China minus its distance to the United States, net of the donor-pool average in the same year, computed separately for human-rights votes and for every other vote where China and the United States voted differently. What the triple difference requires is that these two series move together before treatment, so that any common drift cancels when one domain is differenced against the other.

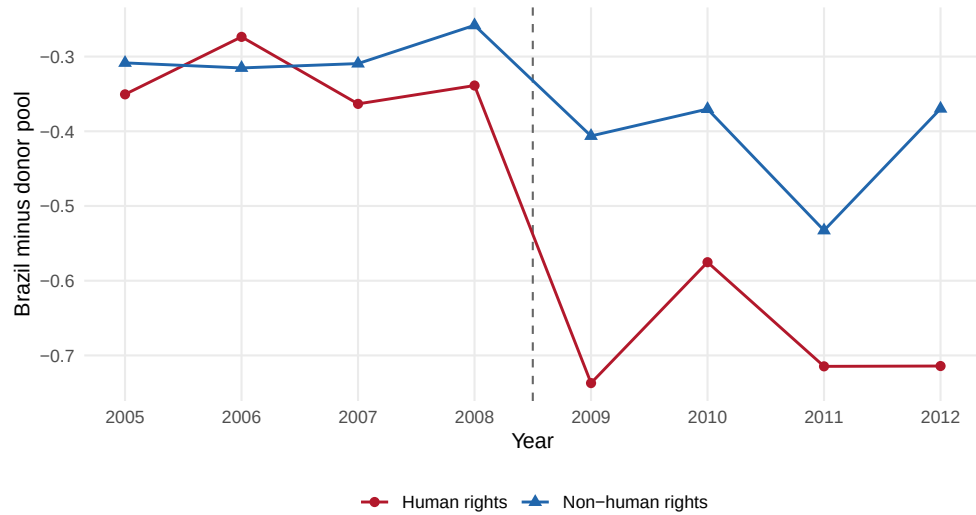


Fig. 8 Brazil's distance from the donor pool in China-US divergent votes, by issue domain. Each point is Brazil's mean vote-level distance to China minus its distance to the United States, minus the donor-pool mean in the same year; more negative values mean Brazil sits closer to China than the donors do. The dashed line marks 2009. The two domains track each other before 2009 and separate afterward, which is the pattern the triple difference assumes and then estimates. Note: Vote-level distance to China minus distance to the United States, relative to the donor-pool mean. Window: 2005-2012 roll-call votes where China and the United States voted differently.

Resolution-Level UNGA Vote Diagnostics

To make the Brazil-China movement in UNGA ideal points more concrete, Table 8 reports the annual share of resolutions in which Brazil and China cast identical votes from 2005 to 2012. The diagnostic uses roll-call votes from `unvotes` 0.3.0, based on Erik Voeten's UNGA voting data, and excludes missing or absent votes by either Brazil or China.

Table 8 Brazil-China identical UNGA votes by year, 2005-2012.

Year	Resolutions	Identical votes	Identical votes (\%)	Mean similarity score
2005	97	80	82.5	0.892
2006	108	88	81.5	0.884
2007	93	78	83.9	0.903
2008	96	73	76.0	0.859
2009	83	72	86.7	0.934
2010	85	74	87.1	0.906
2011	96	83	86.5	0.927
2012	90	71	78.9	0.872

Note:

Roll-call resolutions with valid Brazil and China votes; similarity score ranges from 0 to 1. Window: 2005-2012 annual series.

SDiD

3.0.1 Comparison of DiD, SCM, and SDiD Estimating Equations

To gain intuition for the SDiD estimator, it is useful to compare it with standard DiD and SCM. A DiD model estimates:

$$(\hat{\tau}_{ATT}^{\text{DiD}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min_{\tau_{ATT}, \mu, \alpha, \beta} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \mu - \alpha_i - \beta_t - D_{it}\tau_{ATT})^2$$

Let N_0 and N_1 denote the numbers of control and treated units, and let T_0 and T_1 denote the numbers of pre- and post-treatment periods. Classic DiD corresponds to uniform weighting in this comparison. SDiD instead learns unit weights from pre-treatment outcome histories and assigns optimized weights to pre-treatment periods so that their weighted control-unit outcomes approximate the controls' average post-treatment outcomes. In the full unit-weight vector for each weighted estimator below, controls receive optimized nonnegative donor weights that sum to one, while each treated unit receives the fixed positive weight $1/N_1$. In the full SDiD time-weight vector, optimized nonnegative pre-treatment weights sum to one, while each post-treatment period receives the fixed positive weight $1/T_1$. Treated-post cells therefore enter the objectives with positive weight.

The SCM model can be written as follows:

$$(\hat{\tau}_{ATT}^{\text{SCM}}, \hat{\mu}, \hat{\beta}) = \arg \min_{\tau_{ATT}, \mu, \beta} \sum_{i=1}^N \sum_{t=1}^T \hat{w}_i^{\text{SCM}} (Y_{it} - \mu - \beta_t - D_{it}\tau_{ATT})^2$$

Here $\hat{w}_i^{\text{SCM}}$ denotes the full unit-weight vector: optimized donor weights for controls and the fixed weight $1/N_1$ for each treated unit. SCM uses the donor portion of this vector to match the treated unit's pre-treatment outcome path. SCM can approximate stable unit differences through donor weights when the treated unit lies in the donor convex hull, but it does not add unit fixed effects or optimized time weights. SDiD combines unit weights with unit fixed effects and time weights. The basic SDiD estimating equation is:

$$(\hat{\tau}_{ATT}^{\text{SDiD}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min_{\tau_{ATT}, \mu, \beta, \alpha} \sum_{i=1}^N \sum_{t=1}^T \hat{w}_i^{\text{SDiD}} \hat{\lambda}_t^{\text{SDiD}} (Y_{it} - \mu - \alpha_i - \beta_t - D_{it}\tau_{ATT})^2$$

Note that SDiD retains unit fixed effects α_i (like DiD) while also assigning unit weights $\hat{w}_i^{\text{SDiD}}$ (like SCM) and time weights $\hat{\lambda}_t^{\text{SDiD}}$ (unique to SDiD).

.0.2 SDiD Weights and Donor Pool

Tables 9–16 report the Brazil SDiD diagnostic package behind the compact summary in Table 2. I report the complete donor-weight vector rather than only the largest donors, all pre-treatment time weights, fit and balance diagnostics, rank-based placebo inference, donor-influence checks, window sensitivity, and donor exposure to China’s export expansion.

Table 9: Complete SDiD donor-unit weights for the Brazil synthetic counterfactual.

Rank	ISO3	Donor	Weight	Cumulative weight
1	GEO	Georgia	0.0313	0.0313
2	SGP	Singapore	0.0295	0.0608
3	TCD	Chad	0.0282	0.0890
4	GTM	Guatemala	0.0274	0.1164
5	PRY	Paraguay	0.0272	0.1436
6	BOL	Bolivia	0.0226	0.1662
7	ALB	Albania	0.0221	0.1883
8	COL	Colombia	0.0221	0.2103
9	DOM	Dominican Republic	0.0218	0.2322
10	IND	India	0.0213	0.2534
11	AZE	Azerbaijan	0.0210	0.2744
12	TUR	Turkey	0.0208	0.2953
13	DEU	Germany	0.0208	0.3160
14	MWI	Malawi	0.0182	0.3342
15	CRI	Costa Rica	0.0180	0.3522
16	FJI	Fiji	0.0176	0.3698
17	CYP	Cyprus	0.0168	0.3866
18	ARG	Argentina	0.0164	0.4031
19	PNG	Papua New Guinea	0.0164	0.4194
20	ISL	Iceland	0.0162	0.4357
21	HTI	Haiti	0.0158	0.4514

22	HUN	Hungary	0.0156	0.4670
23	BRB	Barbados	0.0155	0.4825
24	KEN	Kenya	0.0153	0.4978
25	NOR	Norway	0.0151	0.5129
26	SUR	Suriname	0.0150	0.5279
27	TTO	Trinidad & Tobago	0.0146	0.5424
28	DNK	Denmark	0.0141	0.5566
29	LTU	Lithuania	0.0137	0.5703
30	NIC	Nicaragua	0.0137	0.5840
31	HRV	Croatia	0.0137	0.5977
32	POL	Poland	0.0136	0.6113
33	SLV	El Salvador	0.0133	0.6246
34	TUN	Tunisia	0.0132	0.6379
35	EST	Estonia	0.0132	0.6511
36	PAK	Pakistan	0.0129	0.6640
37	JAM	Jamaica	0.0126	0.6765
38	MDV	Maldives	0.0121	0.6886
39	TGO	Togo	0.0118	0.7004
40	LVA	Latvia	0.0114	0.7118
41	ESP	Spain	0.0111	0.7229
42	LBN	Lebanon	0.0109	0.7338
43	SVN	Slovenia	0.0108	0.7446
44	MUS	Mauritius	0.0108	0.7554
45	DZA	Algeria	0.0108	0.7662
46	ROU	Romania	0.0105	0.7767
47	BGR	Bulgaria	0.0105	0.7872
48	GUY	Guyana	0.0103	0.7975
49	CZE	Czechia	0.0101	0.8076
50	AUT	Austria	0.0101	0.8177
51	JOR	Jordan	0.0098	0.8275
52	QAT	Qatar	0.0097	0.8372
53	SWE	Sweden	0.0093	0.8465
54	NAM	Namibia	0.0092	0.8557
55	PAN	Panama	0.0091	0.8648

56	FIN	Finland	0.0089	0.8737
57	NLD	Netherlands	0.0089	0.8826
58	COM	Comoros	0.0089	0.8915
59	SEN	Senegal	0.0088	0.9003
60	ISR	Israel	0.0085	0.9088
61	GRC	Greece	0.0085	0.9173
62	UGA	Uganda	0.0081	0.9254
63	PRT	Portugal	0.0081	0.9335
64	SVK	Slovakia	0.0078	0.9413
65	MDA	Moldova	0.0077	0.9490
66	ITA	Italy	0.0077	0.9567
67	MAR	Morocco	0.0071	0.9638
68	BTN	Bhutan	0.0062	0.9700
69	GBR	United Kingdom	0.0061	0.9760
70	USA	United States	0.0045	0.9806
71	CAN	Canada	0.0040	0.9845
72	LKA	Sri Lanka	0.0040	0.9885
73	BWA	Botswana	0.0036	0.9921
74	IRL	Ireland	0.0032	0.9952
75	LBY	Libya	0.0021	0.9973
76	UKR	Ukraine	0.0009	0.9983
77	FRA	France	0.0009	0.9992
78	ARE	United Arab Emirates	0.0004	0.9996
79	BGD	Bangladesh	0.0004	0.9999
80	EGY	Egypt	0.0000	1.0000
81	MEX	Mexico	0.0000	1.0000
82	BHR	Bahrain	0.0000	1.0000
83	BLR	Belarus	0.0000	1.0000
84	CIV	Côte d'Ivoire	0.0000	1.0000
85	CPV	Cape Verde	0.0000	1.0000
86	ECU	Ecuador	0.0000	1.0000
87	GAB	Gabon	0.0000	1.0000
88	GHA	Ghana	0.0000	1.0000
89	GIN	Guinea	0.0000	1.0000

90	KWT	Kuwait	0.0000	1.0000
91	MDG	Madagascar	0.0000	1.0000
92	NGA	Nigeria	0.0000	1.0000
93	RWA	Rwanda	0.0000	1.0000
94	SWZ	Eswatini	0.0000	1.0000
95	VEN	Venezuela	0.0000	1.0000

Note:

Weights sum to one over the donor pool; Brazil is excluded from the donor pool.

Table 10 SDiD pre-treatment time weights for the preferred Brazil specification.

Year	Time weight	Weight rank	High weight
1997	0.0000	8	No
1998	0.0248	6	No
1999	0.0979	4	No
2000	0.0000	8	No
2001	0.0000	8	No
2002	0.0000	8	No
2003	0.0000	8	No
2004	0.0150	7	No
2005	0.0807	5	No
2006	0.1282	3	Yes
2007	0.3020	2	Yes
2008	0.3515	1	Yes

Note:

The weights are assigned over the 1997-2008 pre-treatment period.

Table 11 Intercept-adjusted pre-treatment fit summary for the preferred Brazil SDiD specification.

Specification	ATT	Placebo SE	Adj. pre RMSPE	Adj. post RMSPE	Post/pre adj. RMSPE	Pre-treatment mean	% of pre-mean
Preferred: no covariates	-0.273	0.131	0.058	0.320	5.544	0.647	-42.2

Note:
ATT is the estimated average causal effect on Brazil after 2009, in absolute UNGA ideal-point distance to China. RMSPE diagnostics subtract the mean pre-treatment gap between Brazil and its synthetic comparison before computing RMSPE, reflecting SDiD's allowance for stable level differences.

Table 12: Pre-treatment outcome balance and fixed values supplied in the preferred SDiD audit.

Variable	Definition	Brazil pre-mean	Synthetic pre-mean	Difference	Std. difference
abs_distance_china	1997-2008 mean	0.647	0.987	-0.340	-0.432

Note:
The preferred specification uses no covariates; these variables are reported as descriptive pre-treatment comparisons only. Synthetic values use the SDiD unit weights and are not residualized synthdid balance statistics.

Table 13 Directional and bilateral placebo-in-space rank inference for the preferred Brazil SDiD estimate.

Comparison set	Directional rank	Directional p	Absolute rank	Bilateral p	Denominator	Note
All valid assignments	3/96	0.031	7/96	0.073	96	Primary rank comparison
Pre-fit RMSPE no larger than twice Brazil	2/70	0.029	4/70	0.057	70	Fit-filter sensitivity; not used to select the main p-value

Note:
The primary distribution includes Brazil and all valid placebo-in-space assignments. The directional rank tests the theory-aligned direction of convergence; the bilateral absolute rank is the conservative two-sided counterpart. Interpreting these ranks as p-values requires sufficiently comparable placebo assignments; the donor exposure audit and the pre-fit filter probe, but cannot guarantee, that condition. The fit-filter row was not used to choose the main result.

Table 14: Sensitivity to influential donors and high-weight donor blocks.

Specification	ATT	Adj. pre RMSPE	Post/pre adj. RMSPE	Removed donors	% change	Inference
Preferred: no covariates	-0.273	0.058	5.544	0	0.0	Main SE and
Drop Georgia (rank 1)	-0.270	0.060	5.318	1	1.1	Point estimate
Drop Singapore (rank 2)	-0.275	0.060	5.427	1	-1.0	Point estimate
Drop Chad (rank 3)	-0.272	0.060	5.339	1	0.3	Point estimate
Drop Guatemala (rank 4)	-0.277	0.059	5.453	1	-1.6	Point estimate

Drop Paraguay (rank 5)	-0.274	0.059	5.412	1	-0.3	Point e
Drop Bolivia (rank 6)	-0.263	0.058	5.423	1	3.6	Point e
Drop Albania (rank 7)	-0.274	0.058	5.537	1	-0.5	Point e
Drop Colombia (rank 8)	-0.270	0.059	5.357	1	1.2	Point e
Drop Dominican Republic (rank 9)	-0.276	0.058	5.592	1	-1.0	Point e
Drop India (rank 10)	-0.273	0.059	5.496	1	-0.2	Point e
Drop top 10 donors by weight	-0.268	0.074	4.395	10	1.6	Point e

Note:

Placebo SEs are not recomputed for donor-deletion rows; those rows are point-estimate and intercept-adjusted fit diagnostics.

Table 15 Sensitivity to plausible time windows around the preferred 1997-2015 specification.

Window	ATT	Adj. pre RMSPE	Post/pre adj. RMSPE	% change	Inference
1997-2013	-0.294	0.056	5.610	-7.8	Point estimate only
1997-2014	-0.320	0.057	6.066	-17.5	Point estimate only
1997-2015 preferred	-0.273	0.058	5.544	0.0	20,000-placebo SE and rank reported in main results
1997-2016	-0.278	0.059	5.624	-1.8	Point estimate only
1998-2015	-0.256	0.039	8.151	6.2	Point estimate only
1999-2015	-0.267	0.027	11.976	1.9	Point estimate only
2000-2015	-0.265	0.028	11.343	2.9	Point estimate only

Note:

The preferred 1997-2015 row uses the placebo SE and rank inference reported in the main results. Other windows are point-estimate and intercept-adjusted fit diagnostics; their placebo distributions were not recomputed.

Table 16 High-weight donor exposure to China in exports of goods and services.

Rank	ISO3	Donor	Weight	China share pre	China share post	Best post rank	Flag
1	GEO	Georgia	0.031	0.006	0.013	14	No top-two China export-destination exposure
2	SGP	Singapore	0.029	0.072	0.112	1	China top export destination post-2009
3	TCD	Chad	0.028	0.038	0.061	2	China top-two export destination post-2009
4	GTM	Guatemala	0.027	0.003	0.007	11	No top-two China export-destination exposure
5	PRY	Paraguay	0.027	0.007	0.006	21	No top-two China export-destination exposure
6	BOL	Bolivia	0.023	0.009	0.033	4	No top-two China export-destination exposure
7	ALB	Albania	0.022	0.012	0.067	2	China top-two export destination post-2009
8	COL	Colombia	0.022	0.008	0.056	2	China top-two export destination post-2009
9	DOM	Dominican Republic	0.022	0.004	0.021	3	No top-two China export-destination exposure
10	IND	India	0.021	0.049	0.060	3	No top-two China export-destination exposure

Note:

High-weight donors are the ten donors with the largest SDiD unit weights. China shares and destination ranks in this table include goods and services. Pre and post refer to 1997–2008 and 2009–2015, respectively; shares are annual averages within each window. Best post rank is China's highest export-destination position attained during 2009–2015. Donor eligibility instead uses goods-only ranks: no donor had China as its largest goods-export destination during 1997–2015. Among these high-weight donors, 1 had China as the top export destination in goods and services during 2009–2015 and 4 were cases where China ranked in the top two, accounting for 10.2 percent of total SDiD weight.

Below I also present the ten largest donor weights used in the main model as a visual summary.

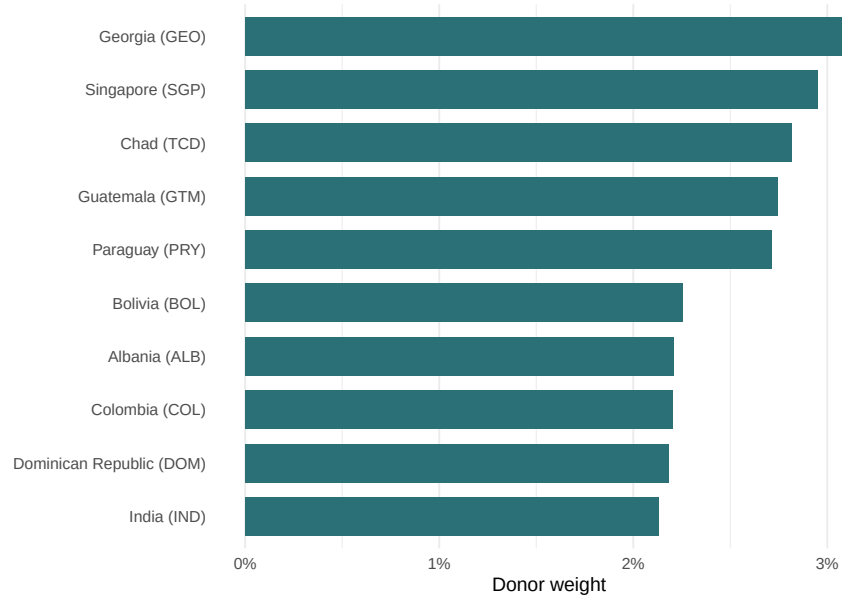


Fig. 9 Ten largest donor weights contributing to the synthetic control in the Brazilian SDiD model. Note: Synthetic-control unit weights, normalized over the full donor pool rather than only over the displayed donors. Window: pre-treatment period ending in 2008.

I also fit a model for a donor pool of Latin American states (excluding Caribbean countries), which is substantively attractive because the donor countries are regionally closer to Brazil. This regional donor pool is a sensitivity check rather than the preferred specification: the main design seeks a pre-treatment synthetic counterfactual with adequate donor support and fit, while the Latin American pool is much smaller.

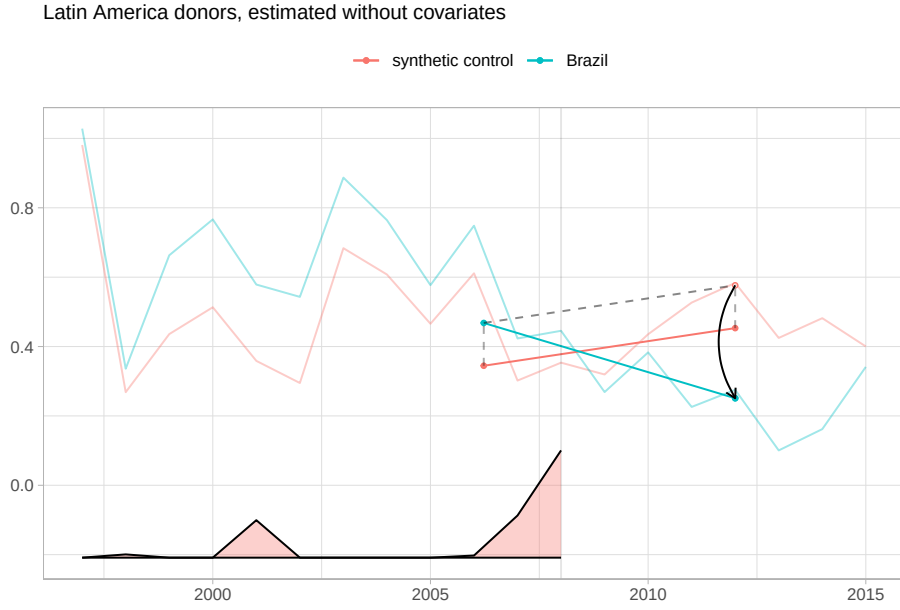


Fig. 10 SDiD fit with a Latin American donor pool, estimated without covariates. Note: Absolute UNGA ideal-point distance to China. SE: 0.207 (placebo, 20,000 replications). Window: 1997-2015 annual series.

Estimated without covariates, the Latin America donor-pool estimate has the same sign and a somewhat larger magnitude: -0.326, compared with -0.273 in the global donor pool. The regional pool has only 19 donors and a slightly higher intercept-adjusted pre-treatment RMSPE (0.069 versus 0.058). Its placebo SE is 0.207, based on 20,000 replications, and the normal-approximation 95 percent interval $[-0.731, 0.080]$ includes zero. Because the directional placebo comparison is computed within this regional pool, which contains only 19 donor units, even Brazil being the 2nd most negative among all 20 assignments yields a p-value of only 0.100. In other words, the regional donor pool does not provide enough power to estimate the effect precisely. The point estimate is nevertheless consistent with the main result and, if anything, is somewhat larger in absolute value than the estimate from the larger global donor pool.

Selected Public-Cue Cross-Country Analyses

This section reports the diagnostics underlying the proposed replacement for the cross-country section. The common country-year window is 1997–2022, and the outcome is absolute UNGA ideal-point distance to China. The individual SDiD models use 105 countries that never have China as their largest goods-export destination during this window. Each individual estimate uses a placebo standard error from 5,000 re-estimations. The pooled models use the interactive fixed-effects estimator in `fet` 2.4.5, with the number of latent factors selected by cross-validation and 1,000 unit-bootstrap replications.

Figure 11 compares the focal cases before their selected public-cue years. Gabon has the smallest mean pre-cue distance to China among the five cases (0.182), which motivates treating its limited room for further convergence as a scope consideration rather than interpreting its exclusion as evidence that no cue occurred.

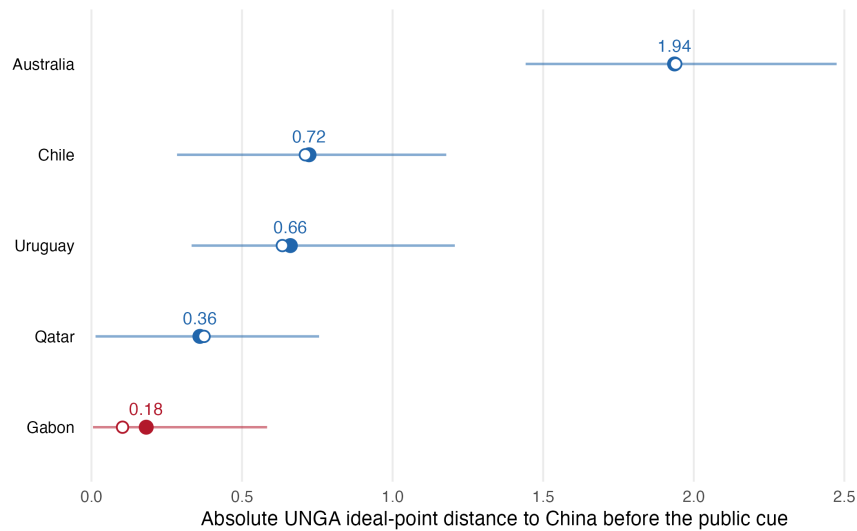


Fig. 11 Pre-cue UNGA ideal-point distance to China among the five focal public-cue cases. Solid points and labels show country-specific means; hollow points show medians; horizontal segments span the observed pre-cue range. Each pre-period runs from 1997 through the year before the selected cue, so the number of years differs across countries. Lower values indicate positions closer to China.

Table 17 reports the individual estimates discussed in the proposed text. Australia 2007 is the theoretically preferred public-cue timing, while Australia 2010 is a robustness timing based on entry into the top goods-export-destination position. For every row, the p-value is the two-sided normal approximation constructed from the placebo standard error; it is not an exact randomization p-value.

Table 17 Individual SDiD estimates for the public-cue cases retained in the proposed cross-country section.

Country	Timing	ATT	SE	95% CI	p-value	Pre/Post	Donors
Chile	Public cue (2008)	-0.130	0.154	[-0.433, 0.173]	0.400	11/15	105
Uruguay	Public cue (2013)	-0.068	0.151	[-0.365, 0.229]	0.652	16/10	105
Australia	Public cue (2007)	0.128	0.161	[-0.187, 0.444]	0.424	10/16	105
Australia	Top goods-export destination (2010)	0.090	0.147	[-0.198, 0.378]	0.540	13/13	105

Note:

ATT is measured in absolute UNGA ideal-point distance to China; negative values indicate convergence. Every row uses a placebo standard error from 5,000 re-estimations. Confidence intervals and two-sided p-values use a normal approximation based on the placebo standard error; the p-values are not exact randomization p-values.

Cross-Country Robustness: Duration Thresholds

The main cross-country estimate uses a five-year durability threshold because the theory concerns status that persists long enough to become politically usable. Table 18 reports the same goods-only IFE design under three- and seven-year thresholds, along with the clean single-entry robustness sample. The estimates remain negative across the restricted-risk-set rows, though precision varies with treated-unit support.

Table 18 Duration-threshold robustness for goods-only China top export-destination status.

Minimum duration	Specification	ATT	SE	95% CI	p-value	Units (T/C)	Treated periods	r*
3	Status-current, restricted risk set	-0.113	0.041	[-0.193, -0.034]	0.005	39/126	477	2
3	Clean single-entry robustness	-0.121	0.041	[-0.202, -0.040]	0.003	26/126	332	2
5	Status-current, restricted risk set	-0.101	0.039	[-0.177, -0.024]	0.010	35/126	440	2
5	Clean single-entry robustness	-0.122	0.041	[-0.202, -0.042]	0.003	25/126	329	2
7	Status-current, restricted risk set	-0.094	0.040	[-0.173, -0.016]	0.019	31/126	411	2
7	Clean single-entry robustness	-0.122	0.042	[-0.205, -0.039]	0.004	23/126	318	2

Note:

ATT in absolute UNGA ideal-point distance to China; lower values indicate convergence toward China. SE: bootstrap with 10,000 replications.

Window: goods-only ITPD-E rank metric. The restricted-risk-set rows retain clean pre-entry non-China-top years and qualifying treated years.

The clean single-entry rows additionally keep only treated countries with one observed qualifying China-top period.

Cross-Country Sample Audit: Goods-Only Treatment Coding

The revised rank metric excludes services before partner ranks are computed. Table 19 documents the ITPD-E broad-sector filter. Table 20 then summarizes how the five-year goods-only rule classifies countries in the UNGA/trade panel.

Table 19 ITPD-E sector filter used to construct the goods-only China top export-destination metric.

Broad sector	Raw rows	Positive-trade rows	Included in rank metric
Agriculture	7331458	2484549	Yes
Manufacturing	71208330	28203707	Yes
Mining and Energy	1205032	416413	Yes
Services	1002558	519520	No

Table 20 Country-level sample audit for the five-year goods-only China top export-destination rule.

Role	Countries
Excluded: no clean observed prior non-China-top year	1
Excluded: pre-2000 China-top entry	2
Excluded: short China-top periods	26
Never-China-top controls	128
Qualifying treated countries	35

Note:

The audit uses the goods-only rank metric and classifies countries before the estimator-specific screen for minimum untreated support. Country counts are therefore a coding audit, not a replacement for the treated/control counts in the estimation table.

Public Cue and Recoverability Audit

To assess how closely the empirical treatment used in the panel regression – China becoming a country’s largest destination for goods exports – maps onto the theorized treatment of a publicly salient status change, I use a source audit originally constructed for 14 countries in an earlier no-covariate treated sample. This audit is not a census of the 35 countries treated in the current goods-only restricted-risk-set specification: 13 audited countries remain in the current treated sample, Qatar does not, and 22 current treated countries were not included in the source audit. The underlying matrix records both recovered and unrecovered China cues. Table 21 displays the nine cases in which the China cue is currently coded as unknown; Chile, Brazil, Uruguay, Saudi Arabia, and Qatar have recovered cues and are omitted by the display filter.

The table distinguishes weak observation from cases in which public rank language about the displaced incumbent is recoverable. For the China cue, `medium` means one countable strong or moderate first-window source with explicit rank language. For the former-incumbent benchmark, `medium` means either one countable source with explicit incumbent-rank or displacement language, or at least two countable trade-coverage records from at least one independent source without a clear top-rank label. A countable source must be eligible, fall within one year of the source-audit entry year, be coded strong or moderate, enter the benchmark, and not be marked `DO_NOT_COUNT`. `Unknown` means that the incumbent was not identified or that no countable benchmark source was recovered; no case is currently coded `low`. Non-recovery is not evidence that a cue was absent because contemporaneous local or official records may be inaccessible or inconsistent with the trade-rank series. Except for Australia, whose displayed year has been reconciled to the current 2009 entry, the table retains the entry windows used in the legacy source audit. Table 22 reports Australia’s broad-trade source separately.

Table 21: Selected unresolved cases from a 14-country legacy source audit of public rank-language recoverability and metric caveats.

Country	Entry year	China cue	Ex-#1 benchmark	Interpretation	Caveat
---------	------------	-----------	-----------------	----------------	--------

Solomon Islands	2003	unknown	unknown	Weak observation/recoverability problem	Additional local/official archive work required.
Philippines	2005	unknown	unknown	Weak observation/recoverability problem	Treatment-rank audit issue in contemporaneous sources.
Angola	2007	unknown	unknown	Weak observation/recoverability problem	Additional local/official archive work required.
Malaysia	2009	unknown	medium	Benchmark recoverable; follow up before recoding salience	Rank-definition and China/Hong Kong aggregation caveat.
Sierra Leone	2012	unknown	unknown	Weak observation/recoverability problem	Local statistics conflict with Belgium-as-incumbent.
Myanmar (Burma)	2014	unknown	unknown	Weak observation/recoverability problem	Additional local/official archive work required.
Saudi Arabia	2013	unknown	unknown	Weak observation/recoverability problem	Corrected 2013-2014 public-cue window remains unresolved; rank-definition reconciliation is required.
Kuwait	2018	unknown	unknown	Weak observation/recoverability problem	Additional local/official archive work required.

Note:

Nine unresolved cases from a 14-country legacy audit; this is not a census of the current treated sample. Definitions and the display filter reported in the text. Australia's entry year follows the current goods-only annual panel; other entry windows remain those of the legacy audit.

Table 22 Australian Financial Review context source for Australia.

Source	Australian Financial Review
Date	2009-11-09
Metric	Australia's largest aggregate trading partner
Counted in export-destination benchmark	No
Raw file	data/raw/ex_top1_salience/AUS/2009/ australian_financial_review/ aus_afr_china_trading_partner_2009.html
Interpretation	Counts for broad public-cue salience in the current 2009 treatment window; remains excluded from the strict export-destination benchmark because the metric combines exports and imports of goods and services.

Note:

The AFR item documents a broad aggregate-trading-partner cue. It documents broad-metric salience during the 2009 annual treatment window. The publication date does not establish that a strict goods-export cue preceded treatment.

ChatGPT Classification

Below is the coded instruction to the ChatGPT API to classify the subjects of the headlines.

```
system_block <- paste(
  paste0(
    'You are a Brazilian expert in International Relations and ',
    'the political'
  ),
  paste0(
    'economy of China-Brazil relations. Classify the subject ',
    'of EACH headline'
  ),
  paste0(
    'when I say EACH, I mean all of them, even if repetitive or ',
    'similar. Do classify all of them'
  ),
  'using one label from this list exactly as written:',
  '"diplomacy", "chinese economy", "china-brasil relations",',
  '"disaster and accidents", "chinese politics and policy",',
  '"sports, science, culture, other", "human rights",',
  '"china-brazil trade", "health".\n\n',
  # 13 static few-shot examples - leave them unchanged!
  'Examples:\n',
  paste0(
    '1. Presidente chinês visita o Brasil e participa de ',
    'reunião com FHC -> china-brasil relations\n'
  ),
  paste0(
    '2. Ministro chinês de Defesa faz visita oficial ao Brasil ',
    '-> china-brasil relations\n'
  ),
  paste0(
    '3. Para governo, novo câmbio chinês só beneficia Brasil no ',
    'médio prazo -> china-brazil trade\n'
  ),
  paste0(
    '4. China ultrapassa EUA como maior parceiro comercial do ',
    'Brasil -> china-brazil trade\n'
  ),
  paste0(
    '5. China vai controlar preços de alimentos e combustíveis ',
    '-> chinese economy\n'
  ),
  paste0(
    '6. Vendas de automóveis na China desaceleram em junho -> ',
    'chinese economy\n'
  ),
  ),
```

```

paste0(
  '7. China quer devolução das relíquias Yin, patrimônio da ',
  'Unesco -> sports, science, culture, other\n'
),
paste0(
  '8. Dinossauro encontrado na China esclarece origem das ',
  'aves -> sports, science, culture, other\n'
),
paste0(
  '9. OMS confirma mais de 4.600 casos de gripe suína; China ',
  'registra 2º caso -> health\n'
),
paste0(
  '10. Marinha chinesa busca "cansar" barcos japoneses em ',
  'águas disputadas -> diplomacy\n'
),
paste0(
  '11. Acidente após casamento deixa 16 mortos na China -> ',
  'disaster and accidents\n'
),
paste0(
  '12. Jornal chinês sai das bancas por foto da praça ',
  'Tiananmen -> human rights\n'
),
paste0(
  '13. China é parceira estratégica contra crise econômica, ',
  'diz UE -> chinese economy'
),
paste0(
  'return as in the examples, a string with the headline and ',
  'the classification'
)
)
)

```

The classified headline file is an archived derived dataset and is not regenerated during manuscript rendering. The preserved producer specifies the OpenAI API model alias `gpt-4.1-mini` with temperature set to zero. It deduplicates the retrieved headlines, retains titles containing `China`, `chinês`, or `chinesa`, divides them into code-defined 500-row slices, opens a new chat object for each slice, splits returned text by line and `->`, removes numeric prefixes, joins classifications back by exact headline text, and normalizes the label string `china-brasil` relations to `china-brazil` relations. These statements document the preserved code configuration, not the exact historical execution that produced every archived

row. The archived file is frozen as the analysis input and records the configured model alias, but the archive does not preserve an immutable model snapshot, a complete request-and-response log, the exact history or rules of manual reruns, or values for generation parameters left at provider defaults. It therefore permits auditing of the stored corpus and code configuration, but not exact re-execution of the historical API calls. The evidence in the main text is used as a headline-level salience diagnostic.

Below I reproduce the 14 archived headline examples and their ChatGPT classifications.

Examples of Classified Headlines

Archived sample: 14 headlines with ChatGPT-assigned topic

Headline	Classification	Date
Disputa com Brasil é 'conflito menor' para China, dizem analistas	china-brazil relations	2005-10-05
Lula vai se encontrar com presidente da China na segunda	china-brazil relations	2004-05-23
China lidera as aquisições de empresas brasileiras	china-brazil trade	2013-03-01
China aceita nova regra brasileira e suspende embargo à soja	china-brazil trade	2004-06-21
PIB da China dobra participação no total mundial em 5 anos	chinese economy	2011-03-25
BHP prevê produção recorde de minério de ferro e aposta na China	chinese economy	2012-01-18
China ameaça Google com "consequências" caso empresa largue autocensura	chinese politics and policy	2010-03-12
China anuncia fechamento de 700 websites	chinese politics and policy	2004-08-01
China não quer interferência dos EUA na questão do Tibete	diplomacy	2011-09-28
Chega a Taiwan primeiro voo direto da China em quase 60 anos	diplomacy	2008-07-04
Incêndio em prédio residencial deixa 12 mortos na China	disaster and accidents	2010-07-19
China detona barragens e prédios danificados; veja vídeo	disaster and accidents	2008-06-06
Uma em três chinesas da zona rural se mata por ano	health	2002-11-29
China suspende viagens ao Tibete para controlar Sars	health	2003-05-13

Fig. 12 Fourteen archived examples of China-related Folha de S.Paulo headlines and ChatGPT classifications. Note: headline text and category labels. Window: sample drawn from the 2000-2014 corpus.

0.3 Validation of ChatGPT Classification

To assess the archived classification, I manually coded a stratified random sample of 100 headlines, allocated in proportion to the sizes of the automated-label categories in the full corpus, with a minimum of five cases per category. The random seed used to select the sample was not retained. Table 23 reports accuracy against manual coding.

Table 23 Accuracy of archived model labels against manual coding in the stratified random validation sample (N = 100). Overall sample accuracy: 88.0 percent.

Predicted label	N	Agreements	Agreement within predicted-label sample (\%)
china-brazil trade	6	6	100.0
disaster and accidents	13	13	100.0
health	6	6	100.0
chinese economy	25	24	96.0
human rights	8	7	87.5
diplomacy	12	10	83.3
sports, science, culture, other	17	14	82.4
china-brazil relations	5	4	80.0
chinese politics and policy	8	4	50.0
Overall	100	88	88.0

Note:

headline-level counts and accuracy by archived predicted label. Window: stratified random validation sample from the 2000-2014 headline corpus.

Overall accuracy in the validation sample is 88.0 percent. The lowest agreement is in the “chinese politics and policy” category (50.0 percent), where several headlines were manually reclassified as “diplomacy.” The category most central to the mechanism, “china-brazil trade,” has 100.0 percent agreement among sampled headlines carrying that automated label. The corresponding rates are 80.0 percent for “china-brazil relations” and 83.3 percent for “diplomacy.” Because the table rows condition on the archived predicted label, the category-specific

rates measure precision rather than recall. This validation does not assess whether classification error changed over 2000–2014. Figure 6 should therefore be read as a descriptive summary of the archived labels subject to classification error.

Cross-Country: Goods-Only Treated Countries

Table 24 describes the 35 treated countries in the main goods-only restricted-risk-set specification. For each country, the table reports the first qualifying China-top goods-export year and the number of treated and untreated country-years retained in the estimation panel.

Table 24 Treated countries in the main goods-only restricted-risk-set cross-country specification.

Country	ISO3c	Treatment year	Treated years	Pre/other untreated years	First panel year	Last panel year
Sudan	SDN	2000	6	10	1990	2005
Solomon Islands	SLB	2003	20	13	1990	2022
South Korea	KOR	2003	20	12	1991	2022
Cuba	CUB	2004	18	14	1990	2022
Oman	OMN	2004	19	13	1990	2022
Philippines	PHL	2005	18	15	1990	2022
Yemen	YEM	2005	11	13	1991	2015
Mauritania	MRT	2006	17	15	1990	2022
Angola	AGO	2007	16	17	1990	2022
Chile	CHL	2007	16	17	1990	2022
Iran	IRN	2007	13	17	1990	2019
Kazakhstan	KAZ	2007	16	15	1992	2022
Congo - Kinshasa	COD	2008	14	18	1990	2022
Japan	JPN	2008	15	18	1990	2022
North Korea	PRK	2008	12	13	1991	2019
Thailand	THA	2008	14	18	1990	2021
Australia	AUS	2009	14	19	1990	2022
Brazil	BRA	2009	14	19	1990	2022
Malaysia	MYS	2009	14	19	1990	2022
South Africa	ZAF	2009	14	15	1994	2022
Peru	PER	2010	13	20	1990	2022
Turkmenistan	TKM	2010	13	18	1992	2022
Congo - Brazzaville	COG	2011	12	18	1990	2022
Zimbabwe	ZWE	2011	7	21	1990	2017
Russia	RUS	2012	11	20	1992	2022
Sierra Leone	SLE	2012	11	22	1990	2022
New Zealand	NZL	2013	10	23	1990	2022
Saudi Arabia	SAU	2013	10	23	1990	2022
Uruguay	URY	2013	10	23	1990	2022
Eritrea	ERI	2014	9	21	1993	2022
Myanmar (Burma)	MMR	2014	9	24	1990	2022
Indonesia	IDN	2016	7	25	1990	2022
Equatorial Guinea	GNQ	2017	6	18	1995	2022
Gabon	GAB	2017	6	27	1990	2022
Kuwait	KWT	2018	5	28	1990	2022

Note:

Ranks are computed from goods exports only. The restricted risk set keeps clean pre-entry non-China-top years and qualifying treated years for treated countries. Treatment year is the first retained year of a qualifying China-top goods-export period.

Measurement Robustness: UNGA-DM Ideal Points

The main estimations in the paper used the classic ideal-point measurement by BSV (Bailey et al. 2017) as the response variable. However, according to Fjelstul et al. (2026), the set of resolutions included in the BSV dataset is not complete. They provided a broader dataset, along with re-estimations of ideal points based on the new data. To assess whether the results of the paper change with this new measurement, I reran the main estimations with this new response variable. To make the comparison as close as possible, I held the treatment definition, donor and risk-set rules, covariate structure, and inference conventions fixed. The UNGA-DM series has a shorter time window, and the SDiD unit and time weights and the IFE factor count are re-estimated by their algorithms in response to the outcome series. These induced adjustments mean that the comparison evaluates the robustness of the analysis package to the alternative outcome; it is not a pure measurement-only perturbation of an otherwise identical fitted counterfactual. It is also important to note that 28 country-session rows had no ISO3c match and 0 SDiD rows had no outcome in the alternative measurement.

The two outcome measures preserve the negative direction of the Brazilian SDiD estimate, but they do not support a blanket claim of robustness across the Brazilian and cross-country designs. Table 25 shows the comparison. Under UNGA-DM, the point estimate is nominally higher than the original in absolute terms (-0.331 versus -0.273), with a lower p-value of 0.008. The directional rank is the same in both measurements (3/96 (0.031)). The bilateral rank improves from 7/96 (0.073) to 5/96 (0.052). The pre-treatment means differ (0.647 under BSV versus 0.837 under UNGA-DM). But the effect measured as a percentage of the pre-treatment average is similar: -42.2 and -39.5 percent. The Brazilian result therefore retains its direction and inferential support under the alternative measure, with a similar relative magnitude. The cross-country comparison below is more sensitive to the outcome source: its IFE point estimate is closer to zero on the UNGA-DM scale and its uncertainty interval includes zero. The 2×2 comparison below holds the common window and country-year panel fixed while varying the outcome source and factor count, allowing conditional factor-count comparisons within each measure. It does not provide a unique, scale-invariant decomposition of the difference between outcomes.

Table 25 Brazil SDiD under two sources of UNGA ideal points. Both columns use the same panel, the same donor pool, and the same placebo standard error settings; the outcome series is the only input that differs, and the synthetic-control weights are re-estimated from it.

	Bailey, Strezhnev and Voeten	Fjelstul, Hug and Kilby
ATT	-0.273	-0.331
Placebo standard error	0.131	0.125
95 percent confidence interval	[-0.529, -0.017]	[-0.576, -0.085]
p (normal approximation, two-sided)	0.037	0.008
Directional placebo rank (p)	3/96 (0.031)	3/96 (0.031)
Two-sided absolute placebo rank (p)	7/96 (0.073)	5/96 (0.052)
Intercept-adjusted pre-treatment RMSPE	0.058	0.058
Brazil pre-treatment outcome mean	0.647	0.837
ATT as percent of pre-treatment mean	-42.2	-39.5

Note:

Estimates are the post-2009 gap in absolute UNGA ideal-point distance to China, 1997-2015 annual panel, no covariates. Placebo SEs use 20,000 replications and seed 20260520 in both columns. Placebo-in-space ranks involve no random draws and are exactly reproducible.

Figure 13 reports the placebo-in-space distribution for the UNGA-DM SDiD fit. It is close to the distribution obtained under the original outcome. The directional placebo rank is unchanged under the alternative measure (3/96 (0.031) in both columns), while the bilateral absolute rank improves from 7/96 (0.073) to 5/96 (0.052). Table 25 reports this cross-measure comparison, and Table 26 reports the donor-exclusion checks under UNGA-DM.

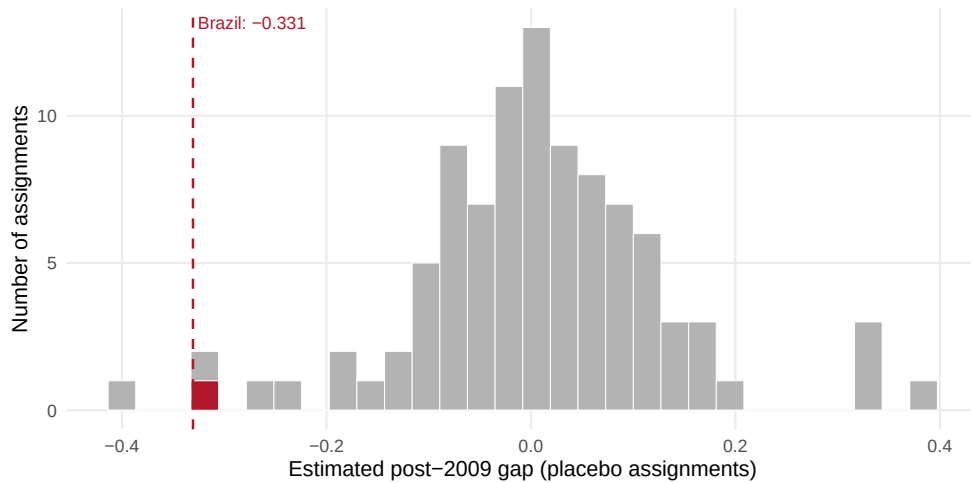


Fig. 13 Placebo-in-space distribution under the UNGA-DM outcome, with Brazil marked. Brazil's estimate is the 3rd most negative of 96 assignments (directional $p = 0.031$); the two-sided absolute rank is 5/96 ($p = 0.052$). Negative values indicate convergence toward China. Note: Estimated post-2009 gap in absolute UNGA ideal-point distance to China, UNGA-DM measure. Window: 1997-2015 annual SDiD panel.

Table 26 Placebo rank inference under the UNGA-DM outcome, by donor-exclusion criterion.

Comparison set	Excluded donors	Directional rank (p)	Two-sided rank (p)
All valid assignments	–	3/96 (0.031)	5/96 (0.052)
Exclude goods-only China-top donor assignments (harmonized: window criterion)	–	3/96 (0.031)	5/96 (0.052)
Exclude goods-only China-top donor assignments (legacy 5-year qualifying criterion; audit only)	GAB, KWT	3/94 (0.032)	5/94 (0.053)

Note:

The window criterion excludes donors observed as China-top in goods exports within 1997-2015, which is the criterion used elsewhere in the paper. The legacy criterion excludes countries that qualify under the five-year duration rule outside the analysis window, and is reported for audit only.

Donor weights are estimated separately for each fit, so they are not identical across outcomes. Table 27 reports how much they change. Weight is spread among donors in a similar way in both fits: 0.044 under UNGA-DM and 0.031 under BSV. The correlation of the donor pool weights is moderate (0.462), but among the top ten donors, 6 are the same. The donor mix therefore shifts moderately across outcomes. For Brazil, the negative estimate and similar relative magnitude are preserved, but the two fits do not use identical counterfactual weights; this comparison does not establish stability of the cross-country result.

Table 27 Ten largest synthetic-control donor weights under the UNGA-DM fit, with the weight the same donor receives under the main outcome.

Donor	UNGA-DM weight	BSV weight
COL	0.044	0.022
IND	0.031	0.021
SEN	0.031	0.009
BRB	0.029	0.015
GTM	0.027	0.027
GHA	0.026	0.000
TTO	0.026	0.015
PRY	0.026	0.027
BOL	0.025	0.023
SGP	0.023	0.029

Note:

Weights are normalized over the full donor pool of 95 units rather than over the displayed donors. The ten donors shown carry 28.8 percent of total weight under the UNGA-DM fit. Across all donors, the correlation between the two weight vectors is 0.462, and 6 of the ten largest donors are shared by the two fits.

Regarding the panel regression, Table 28 shows the IFE estimation in the same time window for both outcomes, and the original estimation with the larger time window. The estimate retains the negative sign but is closer to zero on the UNGA-DM scale, with an uncertainty interval that includes zero: -0.027 with $p = 0.573$. However, the models are not the same, since the number of latent factors chosen by cross-validation is different: 2 under

BSV and 1 under UNGA-DM. The 2×2 comparison below holds the common window and country-year panel fixed while crossing the outcome source with one versus two latent factors, allowing sensitivity to factor count to be assessed within each outcome measure rather than uniquely decomposing the cross-outcome difference.

Table 28 Cross-country interactive fixed-effects panel under the two outcome sources, in a common window with identical rows.

Specification	ATT	SE	95 percent CI	p	Factors (CV)	Country-years
BSV full window 1990-2022 (paper main, stored target)	-0.101	0.039	[-0.177, -0.024]	0.010	2	5,037
BSV common window (≤ 2020 , identical rows)	-0.095	0.040	[-0.173, -0.018]	0.016	2	4,727
UNGA-DM common window (≤ 2020 , identical rows)	-0.027	0.048	[-0.120, 0.066]	0.573	1	4,727

Note:

All rows use the goods-only status-current restricted risk set with 35 treated and 126 control countries. The common-window rows keep exactly the same country-years in both outcome sources. Confidence intervals come from 10,000 bootstrap replications. ATTs and intervals are expressed on each outcome's original scale; raw cross-outcome differences do not identify a scale-invariant measurement contribution.

Table 29 holds the common window and country-year panel fixed and crosses the two outcome sources with one and two latent factors. This 2×2 comparison describes sensitivity to factor count conditional on the outcome measure. Holding the outcome fixed, moving from one to two factors makes the ATT more negative (larger in absolute value): from -0.036 to -0.095 under BSV, and from -0.027 to -0.065 under UNGA-DM. Cross-validation selects 2 factors under BSV and 1 under UNGA-DM. The change from one to two factors differs across the two measures, indicating a descriptive interaction between outcome source and factor count rather than a unique additive decomposition. This comparison does not test the interaction. Both UNGA-DM fits have negative point estimates and uncertainty intervals that include zero; overlapping intervals do not establish equivalence. Holding the factor count at two still leaves a smaller UNGA-DM estimate (-0.065) than the BSV estimate (-0.095), but

these raw estimates are expressed on different outcome scales. Their difference is a conditional sensitivity comparison, not a scale-invariant contribution attributable to outcome source.

Table 29 Outcome source by number of latent factors, holding the panel fixed.

Outcome	Factors (fixed)	Selected by CV	ATT	SE	95 percent CI	p
BSV	1	No	-0.036	0.037	[-0.109, 0.037]	0.336
UNGA-DM	1	Yes	-0.027	0.048	[-0.120, 0.066]	0.573
BSV	2	Yes	-0.095	0.040	[-0.173, -0.018]	0.016
UNGA-DM	2	No	-0.065	0.055	[-0.172, 0.043]	0.238

Note:

Same common window and identical rows as the previous table. Paired bootstrap on UNGA-DM minus BSV in their original outcome units, 1,000 replications and no failures: 0.069 (percentile $p = 0.134$) with the original CV-selected counts held fixed (BSV: two; UNGA-DM: one), and 0.030 (percentile $p = 0.356$) with two factors for both outcomes. The same stratified country draws are used for both measures; factor counts are not reselected within replications. These are scale-dependent contrasts, not a unique decomposition of factor and outcome-source components. Non-rejection does not establish equivalence.

To sum up, the Brazilian SDiD result retains its negative direction and inferential support across the two outcome measurements, with similar relative magnitude but a different absolute scale. The 2×2 IFE comparison shows that two factors produce a more negative ATT under either outcome, with a different within-measure change under BSV and UNGA-DM. Both UNGA-DM fits have negative point estimates and uncertainty intervals that include zero; this does not establish equivalence between them. Even with two factors held fixed, the raw UNGA-DM estimate is closer to zero than the BSV estimate on its own scale. This conditional comparison does not isolate a scale-invariant outcome-source contribution beyond factor selection. The paired test does not detect a statistically significant difference between the ATTs, but this non-rejection does not establish equivalence and may reflect uncertainty.